Porosity impact on TENG performance

"Mesoporosity or microporosity or both of them which of these items have more impact on the TENG application?" *Generated by LeapSpace*

For TENGs, both porosity scales help, but *mesoporosity* is usually the more broadly useful lever; *microporosity* is also valuable, mainly for charge trapping and dielectric polarization, and the strongest designs often combine both. Mesoporous structures repeatedly improve output by increasing effective contact area, surface charge density, and accessible charge-storage space, while microporous features mainly add trap sites, interfacial polarization, and charge-retention pathways. [48] [94] [95] [96] [97] [98] [99] [100] [101] [102] [103] [104] [105] [106] [107] [108]

Which matters more?

Porosity type	Main TENG effect	Strength of impact
Mesoporosity	Increases **effective contact area**, **surface charge density**, **charge storage**, and often **dielectric response**; many porous and mesoporous TENGs show strong output gains when porosity is moderate and ordered. [48] [94] [96] [98] [99] [100] [101] [102] [103] [104] [105] [107] [108]	**Usually larger**
Microporosity	Adds **more charge traps**, **dipole sites**, and **interfacial polarization**, but excessive or poorly controlled microporosity can reduce structural integrity or create leakage/blocked pathways. [95] [106] [109] [110] [111] [112]	**Important, but more conditional**
Both together	Can give the best balance of **contact area + charge trapping + dielectric tuning**, especially when pores are open enough for accessibility but not so large or connected that loss rises. [48] [102] [104] [107] [108] [113]	**Often best overall**

Bottom line

If you must choose one, **mesoporosity has the bigger practical impact for TENG performance** because it more consistently boosts contact area and output across many material systems. [48] [98] [100] [101] [102] [105] **Microporosity is still highly useful, but mainly as a charge-trapping and polarization enhancer rather than the primary driver of output.** [95] [106] [109] [110] [111] **The best TENGs often use both, with controlled mesopores for area and transport plus micropores for trapping.** [48] [104] [107] [108]

Confidence level: **High** — the retrieved studies consistently show porosity improving TENG performance, with mesoporosity more often tied to large output gains and microporosity more often tied to trapping and dielectric fine-tuning.

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